20 (a) Simplify $m^3 \times m^4$

3 + 4 = 7, so $m^3 \times m^4 = m^7$

1 mark
Add powers when multiplying the same base

m^7

(1)

(b) Simplify $(5np^3)^3$

$(5np^3)^3 = 5^3 \times n^3 \times p^9 = 125n^3p^9$

2 marks total
Cube every factor; multiply the powers of p

$125n^3p^9$

(2)

(c) Simplify $\dfrac{32q^9r^4}{4q^3r}$

$32 \div 4 = 8$; $q^9 \div q^3 = q^6$; $r^4 \div r = r^3$

2 marks total
Divide coefficients; subtract powers for each base

$8q^6r^3$

(2)

(Total for Question 20 is 5 marks)

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